



Your company invests **heavily**
in employee training program.

But, **1 in 4** participants is already
looking for another job.

Can Machine Learning **help** prioritize **limited** training seats?

Muhammad Naufal At-Thoriq |



The real problem is **Allocation**



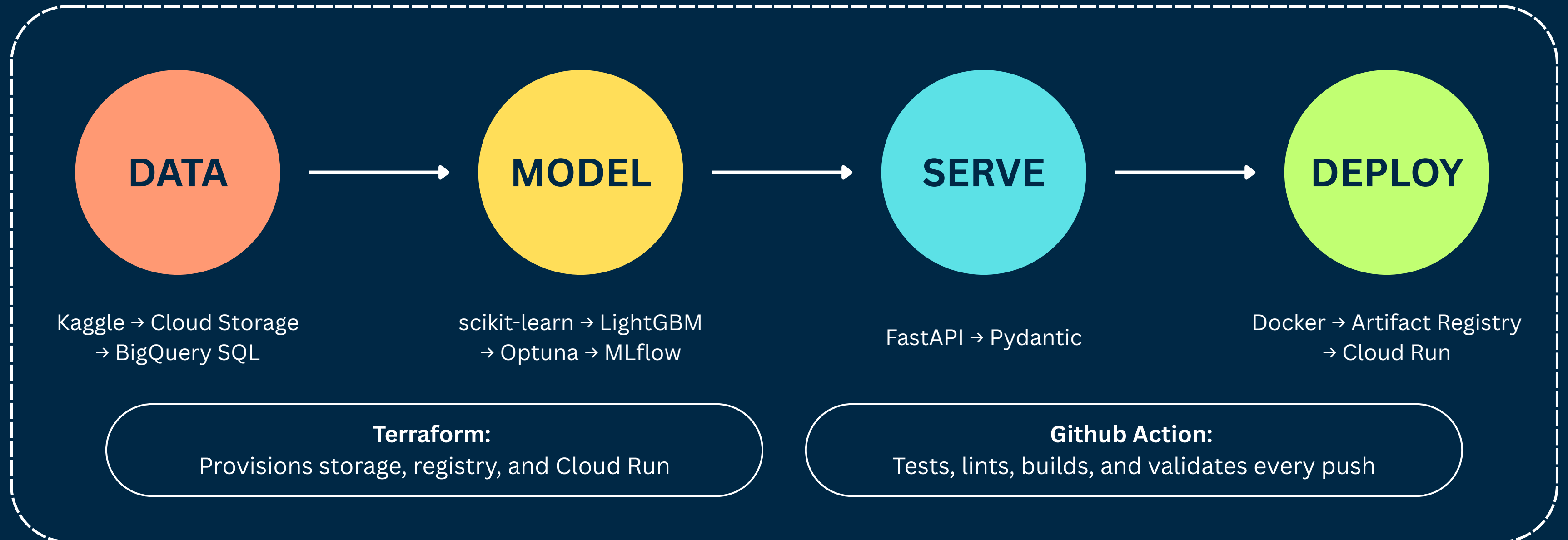
We need to **prioritize** train employee with **lower** job-change intent rate.
Model will **rank** eligible participants **from lower to higher** predicted job-change intent.



Dataset **only** contain Job-Change Intent column; **not future attrition**.
Intent is **not same** as actual resignation. Model prioritize based on **intent**.
Model can be used as decision support; **not guaranteed employment**.

What I built:

A reproducible **ML workflow** from **raw data** to **deployed API**



not just a single-use notebook.

How was the **model selected**?

I compared several **model** with different **preprocessing** and **hyperparameter** config:

Logistic
Regression

Random Forest

LightGBM

LightGBM achieved the **best validation PR-AUC**
across **40 Optuna trials** using **5-fold stratified cross-validation**.



We use **PR-AUC** as main metric since only **±25% data** show **positive** job-change intent.
We also report **ROC-AUC** to evaluate overall ranking performance
and prioritize high **Recall** because false negatives were costlier.

The selected model **improved** ranking

without a large generalization gap



Compared with the dummy baseline, the model **improved PR-AUC** by **~119%** and **ROC-AUC** by **~60%** on the test set.

At the selected threshold, it achieved **~77% recall** on the test set, helping **reduce** costly false negatives.

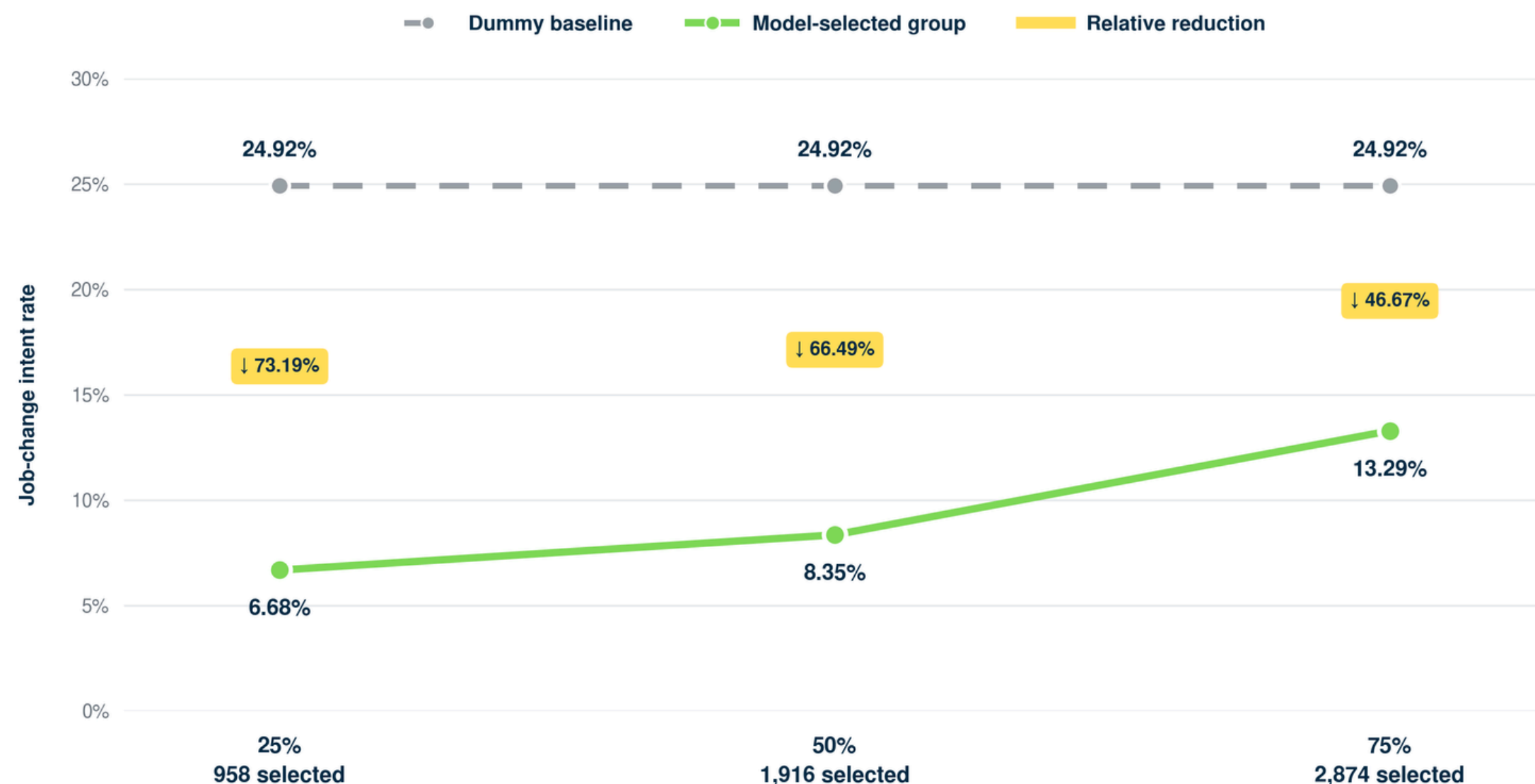
So, does it **help**?

At **every training capacity** level, the model-selected group has a **lower** job-change intent rate than the dummy baseline.

The **strongest** result occurs at **25%** capacity: **958** selected participants have a **6.68%** intent rate, compared with **24.92%** at baseline, a **73.19%** relative reduction.

The advantage **narrows** as more participants are selected, but remains **meaningful** at 75% capacity.

Tldr; Yes, it does help.



Held-out simulation result; not measured production impact

Try the **deployed API** here:

Talent Job-Change Intent Prediction API 0.1.0 OAS 3.1

[/openapi.json](#)

Scores pre-training participants with the complete registered preprocessing and classification pipeline.

default



GET

/health Health



GET

/model-info Model Info



POST

/predict Predict



POST

/predict-batch Predict Batch



[Interactive Documentation](#)



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